

L'Oréal Hackathon

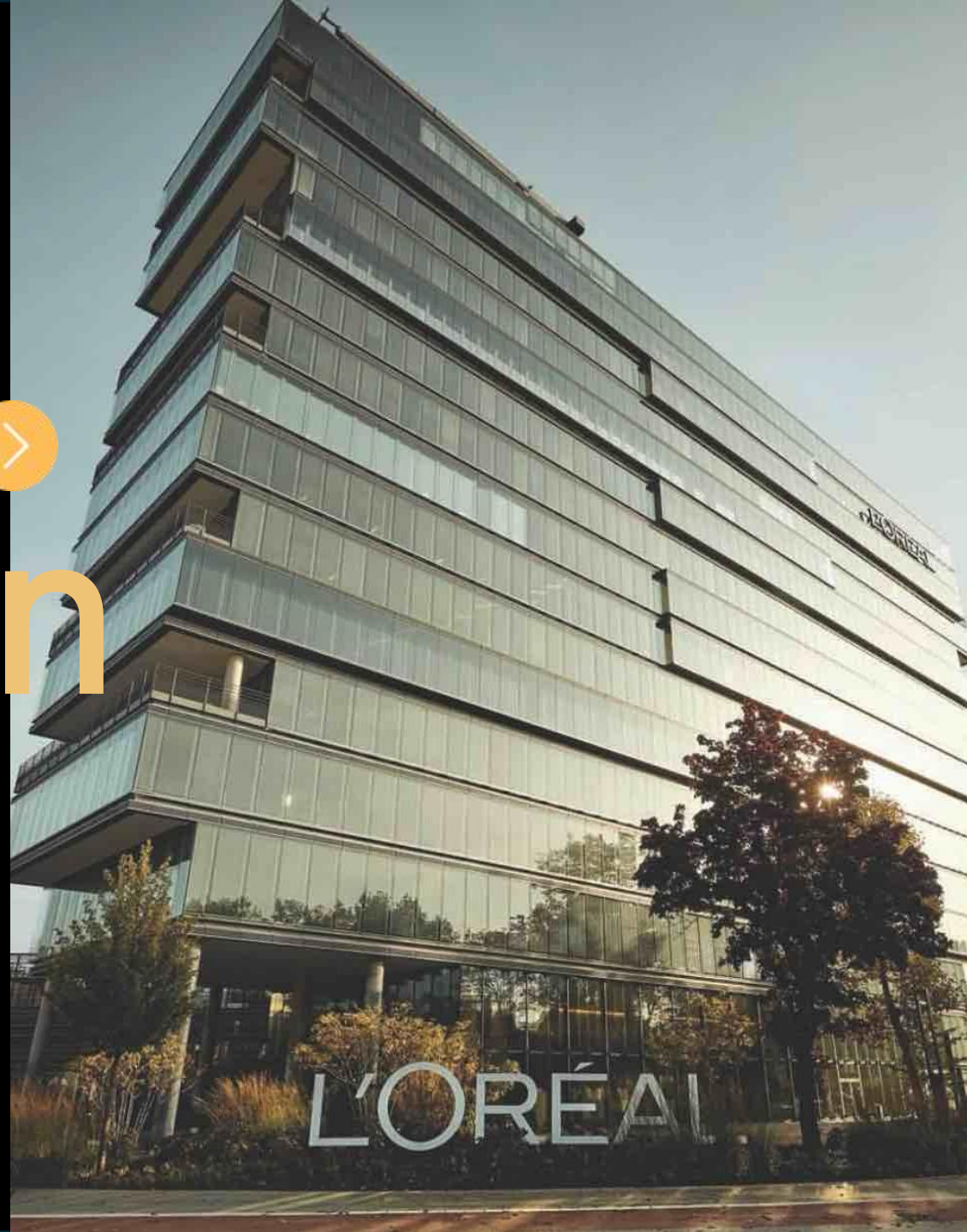
Skin Condition Classification

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Project Details





Overview

The project challenged to develop a sustainable and efficient machine learning model capable of classifying skin conditions based on beauty product descriptions with 33 labels.

Project goals:

- Prediction performance
- Model size
- Resource efficiency

Action standards:

- Performance: Accuracy of F1 score
- Sustainability or evaluation on CO2 emissions

Target Achievement:

- Accuracy of F1 score train target: 70.0-75.00
- Accuracy of F1 score test target: 60.00-60.00
- Sustainability or evaluation on CO2 emissions ≤ 0.100
- Additionally, precision scores within: ≥ 0.50 accumulative

Data Set Information

From the data set given. some of positive label distributions are low in base..

Label Distribution (sorted by sample count):		
Label	Count	Percentage
no_sensitivity	61	0.98%
female	119	1.91%
18-34	195	3.12%
undereye-bags	253	4.05%
55-99	273	4.38%
night	311	4.98%
male	338	5.42%
cleanse	397	6.36%
eye_contour	410	6.57%
sensitivity-high	423	6.78%
oily	469	7.52%
homogeneity	507	8.12%
dark_pigmentation	520	8.33%
sensitivity-low	526	8.43%
day	571	9.15%
combination	579	9.28%
targeted	615	9.86%
normal	719	11.52%
protect	724	11.60%
lack_firmness	762	12.21%
35-54	815	13.06%
dry	884	14.17%
pores	933	14.95%
acne	994	15.93%
eye-wrinkles	995	15.95%
generic	1057	16.94%
lack_radiance	1068	17.12%
wrinkles_fine-lines	1131	18.12%
treat	1167	18.70%
fine_lines	1383	22.16%
care	1390	22.28%
prepare	1885	30.21%
moisturize	2287	36.65%

6240 labelled samples

Label Categories includes:

- **Skin Concerns:** dark pigmentation, acne, wrinkles, etc
- **Skin Types:** dry, oily, normal/combination, sensitive
- **Product Actions:** cleanse, prepare, treat, care, moisturize, protect
- **Demographics:** age-specific, male-targeted
- **Usage:** day, night

Percentile Analysis:

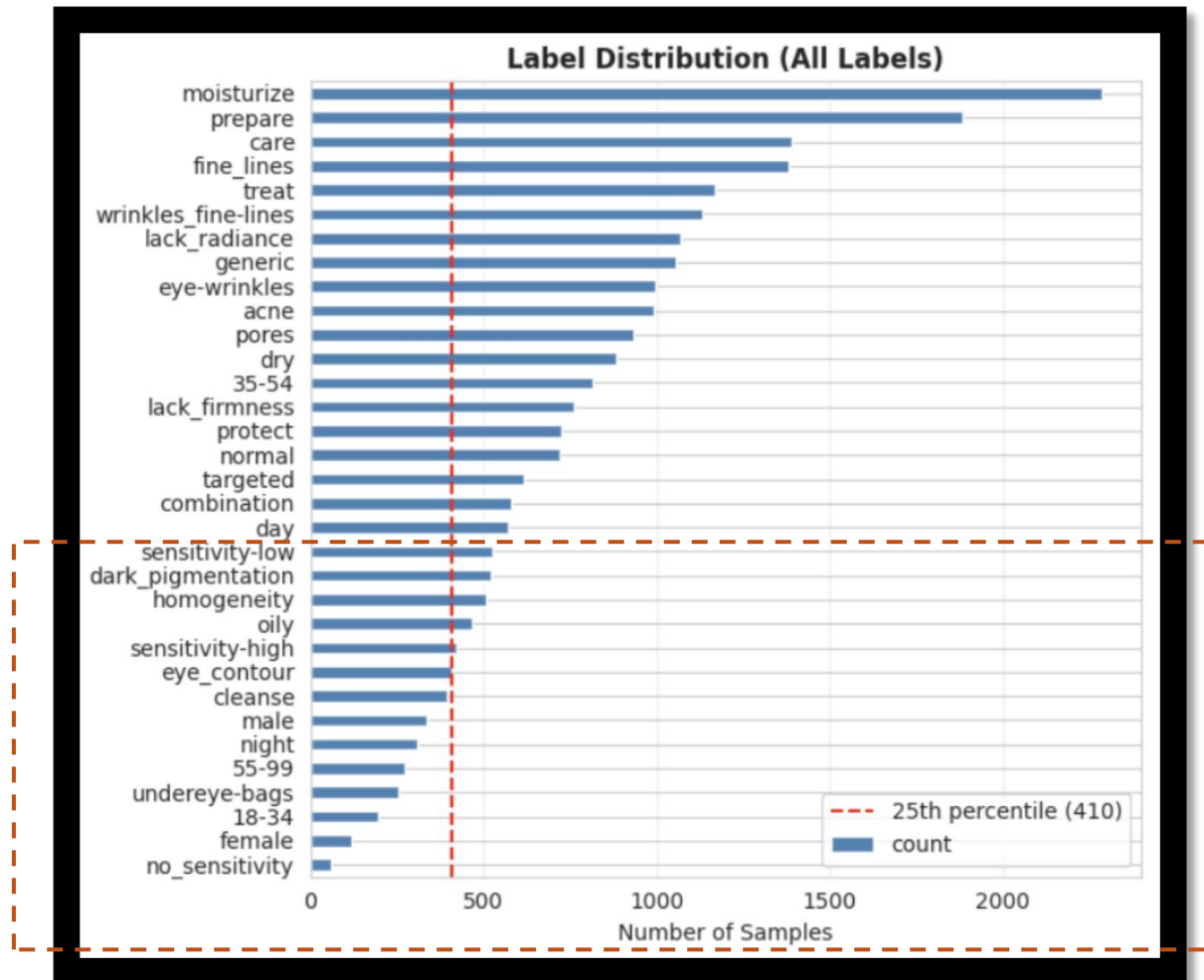
25th percentile: 410 samples

50th percentile: 615 samples

75th percentile: 995 samples

Data Label Distribution

We observe a strong long-tail pattern. We use the percentile threshold as an objective threshold to identify under-represented labels, which motivates our label restructuring strategy.



01

We use the 25th percentile (410 samples) as an objective threshold to identify low-frequency labels for oversampling and restructuring purpose

02

Labels below this threshold are under-represented and may negatively impact model learning.

03

Besides oversampling, we also further explore a few labels overlapping that may create false logic in real-world context setting...

Sustainable ML Strategy

Based on these principles, we implement a data-centric and efficient preprocessing pipeline. These choices reduce computational overhead, improve reproducibility, and support environmentally responsible ML.

01

Data-Centric over Model-Centric

02

Simplicity under Noisy Supervision

03

Efficiency as a First-Class Constraint



What we intentionally avoid:

- Large pre-trained language models
- Extensive hyperparameter search
- Domain-specific handcrafted rules



Executive Summary

01

Model Chosen:

- Ridge Classifier (with default threshold 0)
- F1 Score – Training: 76
- F1 Score – Test: 66
- Precision: 58%

Currently this simple model provides the **robust** scores with only **slight carbon emission**.

02

Improvement Area:

- Threshold confidence level with more data and re-structured labels > if the objective is to increase the precision and more precise recommendation.
- Restructuring and challenge the labels in the future to match with the real-context skincare marketing.
- Train more complex models to increase the reliability of the modelling to predict the product labels.

Pre-modelling Processing

01

Label Restructuring

02

Text Preprocessing

03

Feature Extraction (TF-IDF)

04

Oversampling



Label Restructure Consideration..

Skin Concern: Sensitive

Label	Size Positive Label
Sensitive - High	423
Sensitive - Low	562
Non - Sensitive	61

Suggested to merge into single sensitive label

Skin Concern: Wrinkle

Label	Size Positive Label
Wrinkle_fine-line	1131
Fine-line	1383
Eye-wrinkles	995

1049 overlap

The overlap of fine-line and wrinkle may increase the learning confusing, we merged the label together

Skin Demography: Gender

Label	Size Positive Label
Male	338
Female	119

Low label size, suggested to keep Male and remove female, as the nature of skincare product usually fall into female category

Skin Demography: Age

Label	Size Positive Label
18-34	195
34-54	815
55-59	273

Suggested to merge into single age specified label

Skin Concern: Combination

Label	Size Positive Label
Combination	579
Normal	719

297 overlap

Normal and combination has number of overlap. Therefore, we merged the label together

Label Restructuring

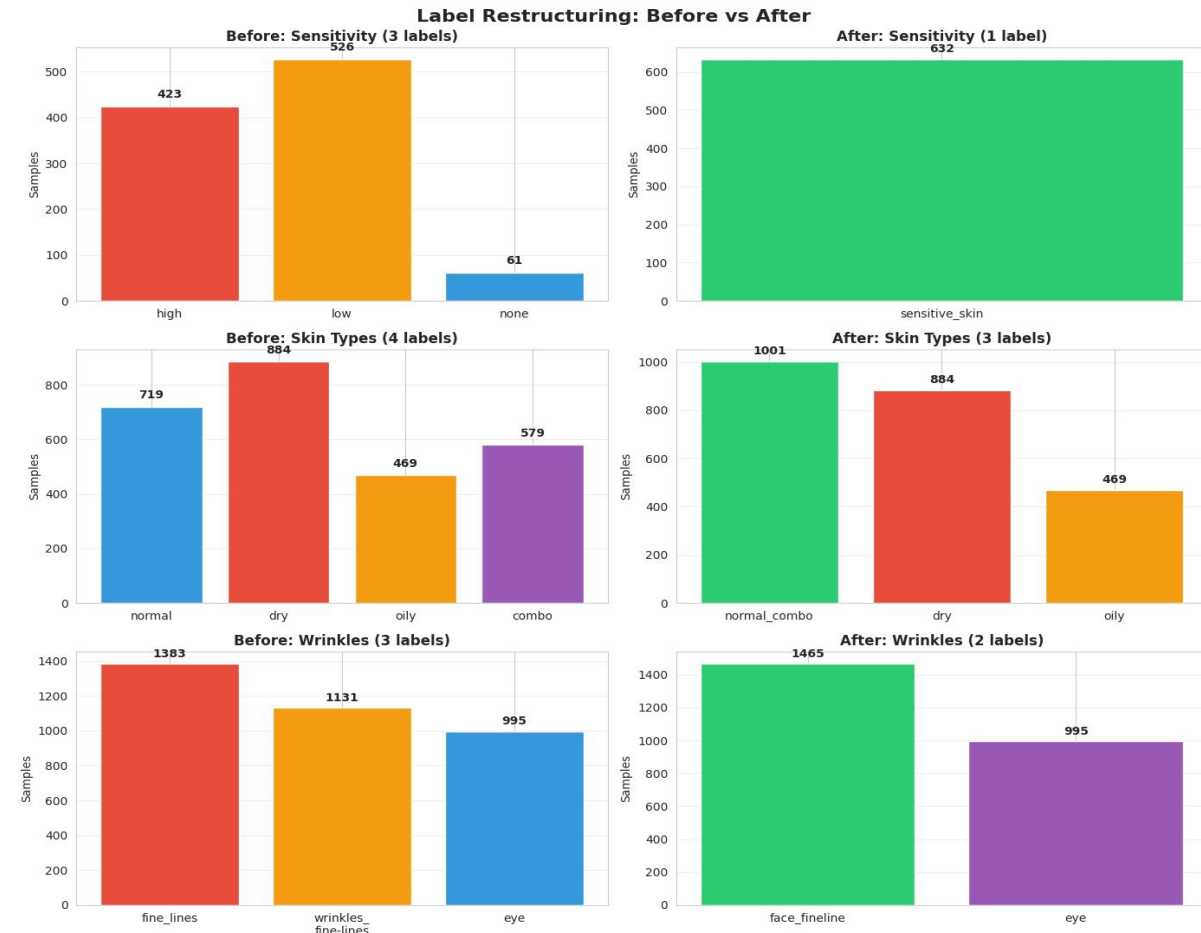
Initial sensitivity labels (high / low / none) show extreme imbalance and ambiguity

Text descriptions lack explicit cues to distinguish fine-grained sensitivity levels

We restructure sensitivity into a single, actionable sensitive_skin label

```
# 1. Sensitivity Labels - Merge high + low:
print("**1. Sensitivity Labels**")
sensitivity_labels = ['sensitivity-high', 'sensitivity-low', 'no_sensitivity']
for label in sensitivity_labels:
    if label in df.columns:
        df[f'{label}_original'] = df[label]
        print(f"    {label}: {df[label].sum()} samples")

df['sensitive_skin'] = ((df['sensitivity-high'] == 1) | (df['sensitivity-low'] == 1)).astype(int)
print(f"    -> Merged into sensitive_skin: {df['sensitive_skin'].sum()} samples")
```



Text Preprocessing

Goal:

preserve semantic signal while avoiding over-engineering

Applied standard and reproducible NLP preprocessing

Lowercasing, tokenization, stop-word removal, lemmatization

No handcrafted rules or domain-specific heuristics



```
sample = df[["text_raw", "text_processed"]].sample(2, random_state=42)
sample
```

...			1 to 2 of 2 entries	Filter		?
index	text_raw	text_processed				
3315	BRUT, After Shave, Solid, Classic scent , 5 oz., (12 Pack) LIVE LIKE A MAN. SMELL LIKE A MAN. Ideal for tightening skin and closing pores after shaving, our Classic Aftershave is not as strong as our cologne, but just as smooth. Simply put, she won't know you're wearing it until she's close enough for it to matter.	brut shave solid classic scent oz 12 pack live like man smell like man ideal tighten skin close pore shave classic aftershav not strong cologn smooth simpli put wont know your wear shes close enough matter				
1760	LOreal Paris Revitalift Moisturizing Anti-ageing Night Cream 50ml Revitalift Moisturizing anti-aging Night Cream	loreal pari revitalift moistur antiag night cream 50ml revitalift moistur antiag night cream				

Show 25 per page

Feature Engineering

Text encoded using TF-IDF with unigrams and bigrams

Effectively captures discriminative marketing phrases (e.g. “sensitive skin”, “anti-aging”)

Produces a sparse and interpretable representation

Well-suited for linear models under noisy supervision

```
# TF-IDF vectorization with unigrams and bigrams:
vectorizer = TfidfVectorizer(
    max_features=5000,
    ngram_range=(1, 2), # Unigrams + Bigrams
    min_df=2,
    max_df=0.95,
    sublinear_tf=True
)

print("Extracting TF-IDF features...")
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

print("Feature extraction complete")
print(f"Feature dimension: {X_train_tfidf.shape[1]}")
print(f"Training matrix shape: {X_train_tfidf.shape}")
print(f"Test matrix shape: {X_test_tfidf.shape}")
print(f"Sparsity: {(1 - X_train_tfidf.nnz / np.prod(X_train_tfidf.shape)) * 100:.2f}%")

*** Extracting TF-IDF features...
Feature extraction complete
Feature dimension: 5000
Training matrix shape: (4992, 5000)
Test matrix shape: (1248, 5000)
Sparsity: 98.60%
```

Oversampling

Goal:

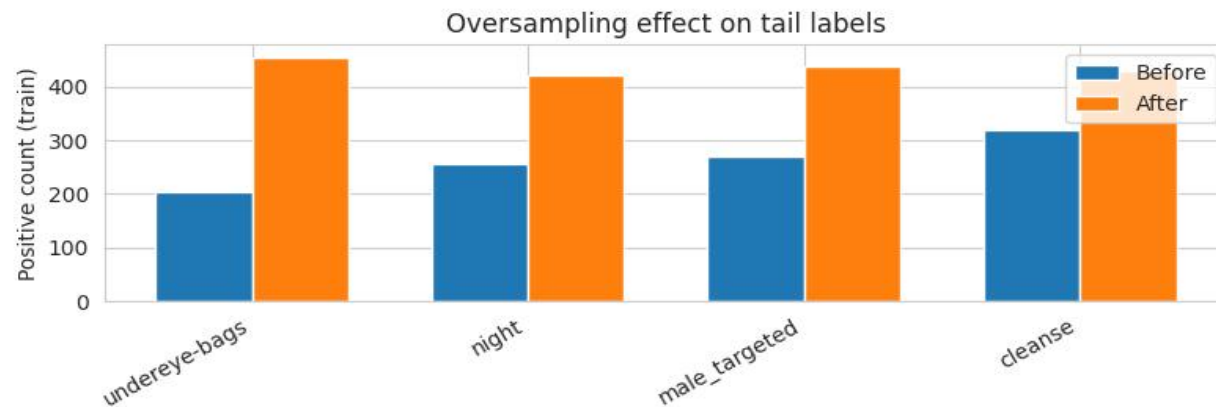
Stabilize learning, not artificially boost metrics

Many labels exhibit extreme class imbalance

Without correction, rare labels are rarely observed during training

Targeted oversampling increases exposure to rare positive examples

Applied only on the training set to stabilize learning without biasing evaluation



Therefore, we use moderate approach to weighting a few of the labels with size of n:400 each

Analysis Insight



Action Standard (1) – F1 Scores

From the models we conducted, Ridge classifier provides satisfied F1 scores for both train and test data, with gap within smaller points

****Model Comparison Results****

Model	Train F1 (weighted)	Test F1 (weighted)	Test F1 (macro)	Test Precision	Test Recall	Gap	Time (s)
Ridge Classifier	0.758769	0.662506	0.653025	0.577565	0.788904	0.096263	0.713252
SGD Classifier (Tuned)	0.686592	0.643624	0.631714	0.548874	0.798343	0.042968	3.431162
Linear SVC	0.703513	0.642623	0.632069	0.552508	0.785221	0.060889	0.667783
Logistic Regression	0.662750	0.621646	0.608330	0.533077	0.767726	0.041104	4.366299

Action Standard (2) – Sustainability

From the models we conducted, Ridge classifier also provides less carbon emission with pretty much robust result given....

****Model Comparison Results****

Model	Train F1 (weighted)	Test F1 (weighted)	Test F1 (macro)	Test Precision	Test Recall	Gap	Time (s)	C02 (kg)
Ridge Classifier	0.758769	0.662506	0.653025	0.577565	0.788904	0.096263	0.713252	0.000002
SGD Classifier (Tuned)	0.686592	0.643624	0.631714	0.548874	0.798343	0.042968	3.431162	0.000008
Linear SVC	0.703513	0.642623	0.632069	0.552508	0.785221	0.060889	0.667783	0.000002
Logistic Regression	0.662750	0.621646	0.608330	0.533077	0.767726	0.041104	4.366299	0.000010

Model Suggested...

From the result given, we choose the Ridge Classifier to further training the model. However, we believe some improvement and construction needed to a few label points in order to increase precision distinction....

BEST MODEL: Ridge Classifier
Train F1 (weighted): 0.758872
Test F1 (weighted): 0.662345
Test F1 (macro): 0.652921
Gap: 0.096526
CO2 emissions: 0.000005 kg

Per-label performance (sorted by F1, worst first):				
Label	F1	Precision	Recall	Support
targeted	0.4030	0.3008	0.6107	131
homogeneity	0.5166	0.4000	0.7290	107
sensitive_skin	0.5284	0.4227	0.7045	132
night	0.5455	0.4286	0.7500	56
care	0.5535	0.4520	0.7138	290
skin_normal_combination	0.5573	0.4441	0.7477	218
prepare	0.5819	0.5158	0.6675	391
cleanser	0.5909	0.4577	0.8333	78
treat	0.6046	0.4971	0.7713	223
skin_dry	0.6052	0.4935	0.7824	193
skin_oily	0.6378	0.5226	0.8182	99
lack_radiance	0.6477	0.5483	0.7910	201
pores	0.6546	0.5747	0.7605	167
dark_pigmentation	0.6667	0.5641	0.8148	108
age_specific	0.6804	0.5731	0.8371	178
day	0.6869	0.5574	0.8947	114
lack_firmness	0.6878	0.5778	0.8497	153
undereye-bags	0.6942	0.5833	0.8571	49
acne	0.7106	0.6345	0.8075	187
male_targeted	0.7333	0.6707	0.8088	68
wrinkle_face_fineline	0.7953	0.7548	0.8404	282
wrinkle_eye	0.7973	0.7458	0.8565	209
moisturize	0.8055	0.7678	0.8471	484
protect	0.8146	0.7444	0.8993	149
eye_contour	0.8235	0.7527	0.9091	77
PROBLEMATIC LABELS (F1 < 0.40)				

Why we believe this approach is performing well enough...



01

Performance +6%: F1 from 0.60 → 0.76
via label restructuring

02

Emissions 0.000065kg →
0.000005kg Speed 15x faster

03

Training 4.67s → 0.79s

04

Scalable: 1.18 MB model size

- *preprocessor.pkl* 1.6 KB
- *vectorizer.pkl* 191 KB
- *best_model.pkl* 987 KB

Further Justification – F1 Scores

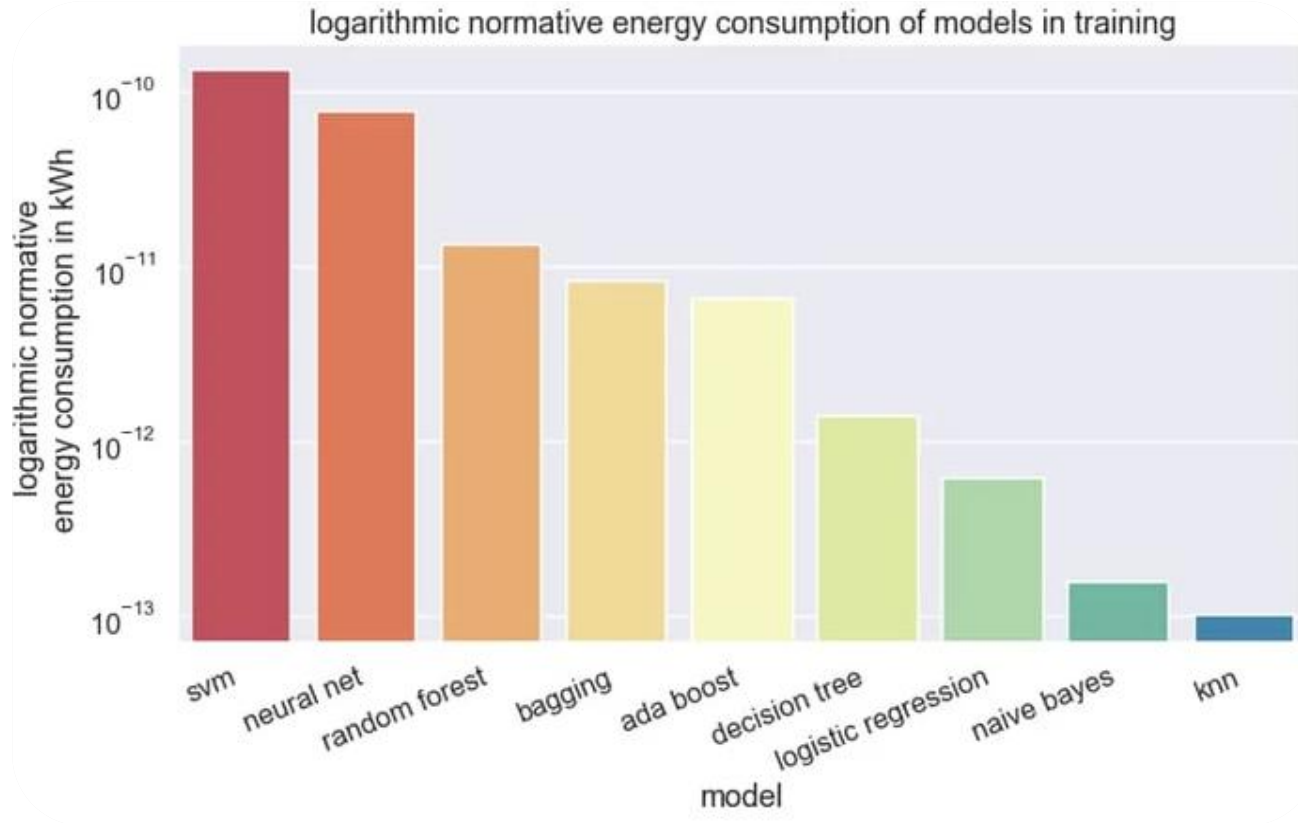
Scenario	Framework	This study (2025)
		$F_1 \uparrow$
Multilabel (Native)	AutoSklearn	0.630–0.816
	AutoKeras	0.248–0.758
	AutoGluon	0.101–0.421
	FEDOT	0.538–0.781
	SMAC	–
	HB	–
	BOHB	–
	GGP	–
	HTN-BF	–
	Random	–

Even without AutoML, our classical multilabel pipeline (TF-IDF + one-vs-rest linear classifier) achieves a test F_1 of 0.66, which falls within the performance ranges reported for native multilabel AutoML frameworks in the benchmark study.

This indicates that a simple, interpretable, and lightweight model can deliver AutoML-comparable performance while being more sustainable and easier to deploy (small artifact, lower compute, simpler training loop).

Aragão, M.V.C., Afonso, A.G., Ferraz, R.C. *et al.* A practical evaluation of AutoML tools for binary, multiclass, and multilabel classification. *Sci Rep* 15, 17682 (2025). <https://doi.org/10.1038/s41598-025-02149-x>

Further Justification – Carbon Emission



Ridge Classifier is an L2-regularized linear model (classification via ridge regression on $\{-1, +1\}$ targets), so its training energy is expected to be in the same low-energy region as Logistic Regression, far below SVM/NN and ensembles.

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.RidgeClassifier.html

<https://www.nextlytics.com/blog/benchmark-of-sustainable-machine-learning-algorithms>



Documentation

Link Google Collabs

https://drive.google.com/file/d/10JiA86Yt7yJ014a8hEqW-UpoE9KSJ_LJ/view?usp=sharing

Link Model Saved

<https://drive.google.com/drive/folders/1gcl3kD-8xtWhg2-NaLF5xJnkvGck5P9g?usp=sharing>

THANK
YOU

